Midterms 10-11 | Formulas & graph rules

Growing reference: Midterms 10-11 only. 2 of 28 modules covered. 11 Nov 2026.

1. Wage and purchasing reward

$w = W/P$

W = money wage/hour; P = goods price. w is pre-tax goods/hour.

$R = [(1 - t_n)/(1 + t_c)] \times w$

R is what an extra work hour can buy after both taxes.

t_n = labor tax; t_c = consumption tax

Use decimal rates: 25% = 0.25. P excludes consumption tax.

$10 \times 0.75 = 7.5$; $10 \div 1.25 = 8$

Separate examples: 25% labor tax versus 25% consumption tax; w held at 10.

SRC-012 slide 26; SRC-083 pp. 2-4. Tax accounting: derived.

2. Individual to aggregate hours

$N_s(w) = \Sigma_i\, h_i(w)$

At one common wage, add each person's offered hours.

$N_s = pop \times h$ (equal hours only)

pop = working-age people; h = hours per person.

$4 \times 5 = 20 \rightarrow 3 \times 5 = 15$

Fewer people; same hours per remaining person.

Reset: $4 \times 5 = 20 \rightarrow 4 \times 4 = 16$

Same people; fewer hours each as leisure becomes more attractive.

SRC-012 slides 21, 26, 35. Sum notation: generated.

3. Read the labor-market graph

Vertical: w = W/P; horizontal: N

N is total labor hours. Demand slopes down; course supply slopes up with lifetime resources fixed.

$N_d(w^*) = N_s(w^*) = N^*$

The intersection gives the market wage w* and labor hours N*.

Same w + fewer supplied hours ⇒ supply LEFT

Wage change alone: move along supply. Supply shifters: lifetime resources (PVLR), taxes, population, value of leisure (VL).

A, K fixed ⇒ labor demand unchanged

A = productivity; K = total capital. Find the new intersection after identifying the supply shift.

SRC-012 slides 19-29; SRC-083 p. 2.

4. Temporary and permanent taxes

Temporary tax ↑: N_s ←; N* ↓; w* ↑; R ↓

Ignore the lifetime-resource effect; hold A, K and other shifters fixed. No offsetting transfer is included.

Permanent tax ↑: SE ←; IE →

SE = lower reward makes leisure relatively cheaper. IE = fewer lifetime resources encourages more work, with normal leisure.

Final N*: lower, unchanged, or higher

The income effect pushes right RELATIVE TO substitution-only supply. Compare the strengths of both effects.

Exact offset: same N*, same w*; R lower

One possible case. The original intersection returns; the tax still applies.

SRC-012 slides 23, 25-27, 32-34; SRC-131; SRC-083 pp. 2-6.

5. Population and leisure shifts

Working-age population ↓ ⇒ aggregate N_s ←

At the same wage, the remaining individual supply schedules can stay unchanged.

Value of leisure ↑ ⇒ individual hours ↓

If this change applies across households, aggregate supply shifts left too.

With demand fixed: N* ↓; w* ↑

Reset between scenarios. Hold taxes, lifetime resources and the other supply shifters fixed for each comparison.

SRC-012 slides 21, 26, 35; SRC-083 pp. 7-8, 11.

6. Output, potential output and LRAS

$Y = A \times K^{0.3} \times N^{0.7}$

Y = real output; N = total labor hours.

$Y^* = A \times K^{0.3} \times (N^*)^{0.7}$

N* = market-clearing hours. Potential is not a physical maximum.

$Y/N = A \times (K/N)^{0.3}$

Output per hour. Fixed A, K: N ↓ ⇒ Y ↓ but Y/N ↑.

LRAS: P vertical; Y horizontal; Y = Y*

P = price level. Flexible W: P ×2 and W ×2 leave W/P, N* and Y* fixed.

A, K or N* change ⇒ LRAS can shift

P alone does not shift LRAS; hold K fixed unless stated.

Sticky wage + hours demanded < hours supplied

Can put actual Y below current Y*. Lower clearing N* alone is not cyclical unemployment.

SRC-012 slides 4, 8, 52-59; SRC-129 p.13.